



Parvathy Shaji kumar

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About me

Computer Science graduate with nearly 4 years of professional experience in Data Engineering and cloud-based data processing. Experienced in designing scalable ETL pipelines, data warehousing solutions, and large-scale data processing using AWS, PySpark, SQL, and Python. Strong academic foundation in Machine Learning, Data Science, and Data Warehousing, complemented by industry experience in analytics and distributed data processing systems. Seeking to pursue a master's degree in data science with interests in Machine Learning, Deep Learning, Big Data Analytics, Computer Vision, and Intelligent Data Processing.

Education & Training

Bachelor of Technology (Computer Science and Engineering) | 2018-2022

University: APJ Abdul Kalam Technological University, Kerala, India

CGPA: 8.74/10

Relevant Coursework: Machine Learning, Data Science, Data Mining and Warehousing, Probability & Statistics, DBMS

Research Interests

Data Science, Machine Learning, Deep Learning, Computer Vision, Big Data Analytics, Data Engineering, Artificial Intelligence, Distributed Computing, Intelligent Data Processing

Work experience

Associate Data Engineer | Cognizant Technology | 02/06/2025 - Present | INFOPARK, Kochi, India

- Developed scalable data ingestion and transformation pipelines using PySpark and AWS Glue for large-scale data processing.
- Designed automated workflows using AWS Lambda, Amazon S3, and ECS Fargate for event-driven data processing.
- Built metadata-driven processing frameworks using DynamoDB for configuration management and auditing.
- Developed monitoring and auditing solutions using EventBridge, SNS, and Lambda to improve pipeline reliability.
- Optimized distributed data processing workflows, reducing execution time for a 42 GB dataset by 75%.

Data Engineer | IBS Software | 13/07/2022 – 31/05/2025 | Technopark, Trivandrum, India

- Developed and maintained ETL pipelines using PySpark, AWS Glue, Python, and SQL.
- Implemented end-to-end data integration pipelines from Amazon S3 to Amazon Redshift.
- Designed and optimized 30+ AWS Glue jobs, reducing ETL processing time and improving resource utilization.
- Built scalable data lakes and data warehouses supporting analytics and reporting workloads.
- Enhanced SQL query performance through optimization techniques and data modeling.
- Created technical documentation for ETL and data warehouse solutions.

Programming Languages: Python, SQL, PySpark

Data Engineering & Big Data: ETL/ELT Pipelines, Data Warehousing, Data Lakes, Data Modeling, Distributed Data Processing, Big Data Analytics, CI/CD Pipelines

Cloud Technologies: AWS Glue, Amazon Redshift, Amazon S3, AWS Lambda, AWS Athena, DynamoDB, AWS Step Functions, EventBridge, ECS Fargate, AWS IAM

Databases: PostgreSQL, MySQL

Tools & Platforms: Databricks, Jupyter Notebook, Git/GitHub, Terraform, VS Code, PyCharm

Soft Skills: Analytical Thinking, Problem Solving, Team Collaboration, Adaptability, Time Management

Academic Project

Deep Fake Detection Using Deep Learning | Final Year B.Tech Project (2021–2022)

- Developed a deep learning framework for detecting deep fake videos.
- Implemented and evaluated XceptionNet and MobileNet architectures for binary classification of authentic and manipulated videos.
- Performed data preprocessing, frame extraction, model training, and model evaluation.
- Explored challenges associated with evolving deep fake generation techniques and adaptive detection methods.

GenAI-Powered Insurance Claim Fraud Detection System | Personal Project (2026)

- Developed an AI-powered insurance claim fraud detection application using Python, Streamlit, and Anthropic Claude.
- Designed a fixed-width file parsing and validation framework for automated insurance claim processing.
- Implemented rule-based and fuzzy matching techniques to identify duplicate and suspicious claims.
- Built an explainable fraud detection workflow that generated fraud indicators, similarity scores, and recommended actions.
- Developed an interactive Streamlit dashboard for fraud analysis and reporting.
- Applied concepts from Generative AI, data validation, fraud analytics, and intelligent decision support systems.

International Experience

All Nippon Airways (ANA), Japan – Client-Side Assignment | 2024

- Worked onsite with ANA teams during a 3-month User Acceptance Testing (UAT) phase.
- Supported testing, validation, and issue resolution activities to ensure successful project delivery.
- Collaborated with international stakeholders and coordinated communication between onsite and offshore teams.
- Gained experience working in a multicultural business environment and addressing real-world client requirements.

Certifications

- **Databricks Certified Data Engineer Associate – Databricks**
- **AWS Certified Cloud Practitioner – Amazon Web Services**

Language Skills

- **English:** Fluent (IELTS Academic: 7.0)
- **German:** Basic (A2 level)